Asim Mohamed

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Education

African Institute for Mathematical Sciences (AIMS) Feb. 2024 – November 2025
Master's Degree, Machine Intelligence

- **Thesis Supervisor:** Dr. Martin Gubri - Research Lead at Parameter Lab
- **Relevant Coursework:** Computational Linear Algebra, Mathematics for Machine Learning, Foundations of Machine Learning and Deep Learning, Computer Vision, Optimization for Machine Learning, Natural Language Processing, Machine Learning Operations.

University of Khartoum Oct. 2015 – Dec. 2021
B.Sc. (Hons) Electrical and Electronic Engineering (Electronic Systems Software Engineering.)

- **CGPA**: 7.93/10 (First Class Honours Degree)
- **Thesis Supervisor:** Dr. Anas Showk - Associate Professor at the University of Khartoum
- **Relevant Coursework:** Data Structures and Algorithms, Database Systems, Java Programming and Applications, Neural Networks and Fuzzy Logic, Distributed and Parallel Computing, Computer Networks, AI and Expert Systems.

Experience

AI Engineer, Enigma-AI April 2025 – Present

- Designed and deployed production-ready AI agents integrating multiple large language models through a unified orchestration framework.
- Designed and deployed enterprise-scale RAG pipeline with hybrid retrieval strategy combining dense embeddings and sparse lexical search, improving retrieval precision by 40%.
- Deployed multi-tenant AI platform with organization-level isolation, role-based access control, and comprehensive security boundaries for enterprise customers.

Software Engineer, Enigma-AI Feb. 2023 – Jan. 2024

- Developed customizable and user-friendly chatbots enabling businesses to tailor interactions to meet specific customer needs, using technologies such as Python, Node.js, and the MERN stack.
- Worked closely with cross-functional teams to design user-friendly chatbot interfaces and integrate them with platforms like Telegram, WhatsApp, and Messenger.
- Led the redesign and enhancement of the Enigma website and platform, improving both user experience and functionality with React.js and TypeScript.

Full-Stack Engineer, Malomatia May 2022 – April 2023

- Contributed to the Supreme Judiciary Council (SJC) E-service team in Qatar, primarily working as a React.js and ASP.NET developer. Focused on modernizing the existing software by transitioning from ASP.NET Webforms to React.js with TypeScript to deliver a more efficient and up-to-date solution.

Research Publications

Is Multilingual LLM Watermarking Truly Multilingual? A Simple Back-Translation Solution October 2025
Under Review [Paper Link]

Enhancing Arabic Readability Prediction with Hybrid BERT and Linguistic Features August 2025
SIGARAB ArabicNLP 2025 BAREC Shared Task Submission [Paper Link]

From Nile Sands to Digital Hands: Machine Translation of Coptic Texts August 2024
The Second Arabic Natural Language Processing Conference [Paper Link]

Teaching Experience

White Nile University, Lab Instructor Oct. 2023 – Jan. 2024

- **Courses**: Algorithms and Data Structures, Multimedia Technology and Applications.

University of Khartoum, Lab Instructor Sep. 2022 – Dec. 2022

- **Courses**: Multimedia Technology and Applications, Networked Multimedia.

Conferences / Summer Schools

MenaML Winter School 2026, KAUST – Saudi Arabia — January 2026 [Certificate]
Deep Learning Indaba 2025, Kigali – Rwanda — August 2025 [Certificate]
Cohere Labs Open Science Community Summer School — July 2025 [Certificate]
Deep Learning Indaba 2024, Dakar – Senegal — September 2024 [Certificate]

Presentations

Poster Presentation at Deep Learning Indaba 2024, Dakar – Senegal — September 2024

- **Title**: From Nile Sands to Digital Hands: Machine Translation of Coptic Texts.
- **Summary**: Presented research on building the first Coptic-centered corpus and evaluating the performance of large language models (LLMs) in translating between Coptic, English, and Arabic. [Poster Link]

LLMs Practical Workshop at IndabaX Sudan 2025, Remote — December 2025

- **Summary**: Presented a practical workshop on foundational LLM concepts, covering essential topics such as tokenization, embeddings, and the attention mechanism in transformer architectures. [Workshop Link]

Awards

- Jan. 2024 - **African Master's in Machine Intelligence (AMMI)**
  Awarded a fully funded one-year intensive graduate program (sponsored by Google and Meta AI) that provides brilliant young Africans with state-of-the-art training in machine learning and its applications.

Projects

Towards Robust and Safe AI: a Comprehensive Survey — *In progress*

- Conducted an in-depth review of adversarial vulnerabilities in multilingual and low-resource language models, classifying attack methods and analyzing their implications for model safety and reliability.
- Examined the impact of prompt injection attacks on the robustness of multilingual LLMs, focusing on their ability to bypass safety mechanisms and generate harmful or unintended outputs under adversarial conditions.

Prediction of Regional Biocapacity Using Satellite Imagery and Deep Learning — *Undergraduate Thesis Project*

- Conducted a comparative analysis of techniques for utilizing multi-spectral satellite imagery data.
- Introduced a methodology for assessing biocapacity in Tuti Island and Port Sudan cities. [Github Link]

Cassava Disease Classification

- Developed a convolutional neural network using a self-supervised approach to classify fine-grained cassava plant leaf images into four disease categories or as healthy leaves. [Github Link]

Skills

Programming Languages: Python, JavaScript, TypeScript, C#, Dart, SQL, Java
ML & DL: TensorFlow and Keras, PyTorch, Jax, Flax, Numpy, scikit-learn, Pandas, OpenCV, Airflow, Matplotlib, Seaborn, MATLAB, HuggingFace, LangGraph, LangChain, Transformers, Vector Databases, AI Agents, Retrieval-Augmented Generation
Tools: Git, GitHub, Google Cloud, AWS, MongoDB, SQL developer tool, PostgreSQL, Docker
Frameworks: React.js, FastAPI, Flutter, ASP.Net Core, Node.js, Nest.js, Flask

Communities

Cohere Labs Open Science Community | Masakhane | Indaba Community | Sudanese Machine Learning Community | EleutherAI

References

- Dr. Martin Gubri: Email | Google Scholar | Research Lead at Parameter Lab
- Laurens van der Maaten: Email | Google Scholar | Member of Technical Staff at Anthropic
- Dr. Anas Showk: Email | Google Scholar | Associate Professor at the University of Khartoum